13 The size of the largest angle in a triangle is 4 times the size of the smallest angle.
The other angle is 27° less than the largest angle.

Work out, in degrees, the size of each angle in the triangle.
You must show your working.

Let the smallest angle be $x°$.

Angles: x, $4x$ and $4x - 27$

1 mark
Express at least two angles

$$x + 4x + 4x - 27 = 180$$

1 mark
Angles in a triangle sum to 180°

$$9x - 27 = 180$$

(+27) (+27)

1 mark
Collect like terms

$$9x = 207$$

(÷9) (÷9)

1 mark
Solve the equation

$x = 23$; $4x = 92$; $4x - 27 = 65$

1 mark
All three correct angles

23 °, 65 °, 92 °

(Total for Question 13 is 5 marks)